

12.775

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Question

Choose solution set S corresponding to the systems of two equations

$$x - 2y + z = 0$$

$$x - z = 0$$

① $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$

② $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \mid \alpha, \beta \in \mathbb{R} \right\}$

③ $S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \mid \alpha, \beta \in \mathbb{R} \right\}$

④ $S = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\}$

Theoretical Solution

The solution set S corresponds to the points lying on the intersection of two planes.

The parameters of the given planes are

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \mathbf{n}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, c_1 = 0, c_2 = 0. \quad (1)$$

The direction vector of the line of intersection is given by

$$\mathbf{m} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (2)$$

$$\mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix} \left| \begin{array}{cc} \mathbf{n}_{1_{23}} & \mathbf{n}_{2_{23}} \\ \mathbf{n}_{1_{31}} & \mathbf{n}_{2_{31}} \end{array} \right| \\ \left| \begin{array}{cc} \mathbf{n}_{1_{31}} & \mathbf{n}_{2_{31}} \end{array} \right| \\ \left| \begin{array}{cc} \mathbf{n}_{1_{12}} & \mathbf{n}_{2_{12}} \end{array} \right| \end{pmatrix} \quad (3)$$

Theoretical Solution

$$\left| \mathbf{n}_{1_{23}} \quad \mathbf{n}_{2_{23}} \right| = \begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix} = 2 \quad (4)$$

$$\left| \mathbf{n}_{1_{31}} \quad \mathbf{n}_{2_{31}} \right| = \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2 \quad (5)$$

$$\left| \mathbf{n}_{1_{12}} \quad \mathbf{n}_{2_{12}} \right| = \begin{vmatrix} 1 & 1 \\ -2 & 0 \end{vmatrix} = 2 \quad (6)$$

$$\mathbf{m} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (7)$$

Theoretical Solution

The equation of the line is given by,

$$\mathbf{x} = \mathbf{h} + \alpha \mathbf{m} \quad (8)$$

here

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (9)$$

$$\implies \mathbf{x} = \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (10)$$

The solution set S consists of all points $\mathbf{x}$ on the line:

$$S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\} \quad (11)$$

Plot

